INCH-POUND

MIL-C-70568B (AR)
06 February 1992
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, INPUT/OUTPUT: 9360881

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Fire Control Processor input/output circuit card assembly referred to as the Input/Output Board (IOB). The IOB is a 7 by 10 1/2 inch, double-layered, multi-wire circuit card assembly mounted vertically in the fire control processor (FCP). The IOB performs almost all input and output functions for the FCP in the M163A2 and M167A2 Vulcan Air Defense Systems. Refer to figure 1 for the IOB functional block diagram (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360881	-	Input/Output Assembly
9360883	-	Input/Output Schematic
9361335	-	Test Program Set Manual
9361380	-	IOB Interface Device
9397497	-	Cable Assembly W1
9397498	-	Test Plug P3
9397499	-	Test Plug P5
9397500	-	Test Plug P4

SPECIAL PACKAGING INSTRUCTION

PS 9360881	-	Packaging of Input/Output Assembly

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425 - Design and End Product Requirements for Discrete Wiring Boards

ELECTRONIC INDUSTRIES ASSOCIATION

RS-422 - Electrical Characteristic of Balanced Voltage Digital Interface Circuits

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360881, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the IOB shall meet the requirements of MIL-F-13926 and ANSI/IPC-DW-425 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.2.2 ANSI/IPC-DW-425. IPC-DW-425/11 Type C1 plated through connection.

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3.3 Environmental service conditions.

3.3.1 Vibration. The IOB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 Hertz	0.06 inches double amplitude
15 to 25 Hertz	0.04 inches double amplitude
25 to 55 Hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 Hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 Shock. The IOB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The IOB shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-Operating. The IOB shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The IOB shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Non-operating (storage). The IOB shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The IOB shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

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3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram (9360883) of the IOB.

3.5 Performance.

3.5.1 Functional description. The IOB interface is used by all major software modules except the Servo Control Processor (SCP) subprogram, which does not have access to this interface. The IOB is the primary interface between the FCP and the A2 control panel and remote BIT cards. The IOB shall maintain interfaces between the FCP and the following assemblies.

3.5.1.1 A2 control panel. The FCP shall synchronize data request commands, from the A2 control panel through a dedicated universal asynchronous receiver/transmitter (UART U54) that interfaces with the A2 control panel remote BIT assembly. All data transmitted over the UART link shall be in serial digital format. Data received and transmitted by the FCP through UART U54 shall include all of the input and output functions associated with the BALLISTICS/BIT panel. Data transmitted directly (non-UART) from the FCP through the IOB to the A2 control panel remote BIT assembly shall include the SCP action switch lockout relay computer power lamp and fault lamp illumination commands. Data received directly by the IOB from the A2 control panel remote BIT assembly shall include the status of the VADS mode and external range enable switches, the radar-radiate foot switch position, the gun-fire trigger and action switch positions and the positions of the fault-bypass and acquisition and tracking (ACQ/TRK) switches. The manual speed analog signal is also received directly by the IOB from the A2 control panel.

3.5.1.2 Remote BIT assemblies. The IOB shall communicate with each of the other remote BIT assemblies by means of two UARTs (U42 and U46). The IOB shall interface with remote BIT assemblies located in the radar Unit 2 receiver-transmitter, the radar Unit 5 power supply, the radar Unit 4 range computer and the A1 distribution box. The Unit 4 remote BIT assembly shall automatically transmit range data on dedicated UART U42 at approximately 250 12-bit range updates per second. Radar coast, track, jam, and search status data shall also be transmitted to the fire control processor by U42. UART U42 shall also be used for Unit 4 BIT mode and test switch selection and receipt of BIT data. The input and output data lines for U46 shall be multiplexed so that U46 can interface with any one of the other three BIT assembled. UART U46 shall be used primarily during BIT operations.

3.5.1.3 M61A1 sight. The IOB shall be interfaced directly to the M61A1 sight. FCP data transmitted to the sight shall consist of the READY-TO-FIRE lamp command, the TRACK/JAM/RADIATE lamp command and the analog range data for the sight range meter.

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3.5.2 Power requirements. The IOB power requirements shall be as specified in Table I.

TABLE I. Power requirements.

PARAMETER	+5 Vdc	+15 Vdc	-15Vdc
VOLTAGE TOLERANCE	± 0.25 V	± 0.75 V	± 0.75 V
CURRENT	3 A max	75 mA max	50 mA max
RIPPLE	150 mV P-P max	450 mV P-P max	450 mV P-P max
INPUT	P1-32-34,72-74	P1-40,80	P1-39,79
INPUT RETURN	P1-9, 18, 35-37, 49, 58, 75-77	P1-38, 78	P1-38, 78

3.5.3 Voltage detector circuit. With all power supply voltages applied to the IOB per 3.5.2, COMPUTER POWER LAMP at P2-20 shall be logic low. When one or more of the supply voltages fall below its limit stated below, P2-20 shall be logic high:

Supply	Limit
+5 V	+4.5 ± 0.3 V
+15 V	+13.0 ± 0.9 V
-15 V	-13.5 ± 0.4 V

COMPUTER POWER LAMP (P2-20) shall be capable of sinking 50 milliamperes maximum at 24 ± 2 Vdc.

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3.5.4 Clock circuit. With power applied to the IOB per 3.5.2 and a 500.000 KHz ± 74 Hz TTL square wave clock signal applied to P1-50, clock signals shall be available as differential RS-422 outputs at the following locations:

Connector-Pin	Output Signal
P2-12	UNIT 4 CLOCK (500 KHz) ± 74 Hz HI
P2-47	UNIT 4 CLOCK (500 KHz) ± 74 Hz LO
P2-16	UNIT 2 CLOCK (500 KHz) ± 74 Hz HI
P2-51	UNIT 2 CLOCK (500 KHz) ± 74 Hz LO
P2-15	UNIT 5 CLOCK (500 KHz) ± 74 Hz HI
P2-50	UNIT 5 CLOCK (500 KHz) ± 74 Hz LO
P2-14	DISTRIBUTION BOX CLOCK (500 KHz) ± 74 Hz HI
P2-49	DISTRIBUTION BOX CLOCK (500 KHz) ± 74 Hz LO
P2-13	CONTROL PANEL CLOCK (500 KHz) ± 74 Hz HI
P2-48	CONTROL PANEL CLOCK (500 KHz) ± 74 Hz LO

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3.5.5 Control Circuits. With power applied to the IOB per 3.5.2, applying each IOB input control signal group shown in the left-hand column of Table II shall activate the corresponding control signal shown in the right-hand column of Table II.

TABLE II. Control circuit logic.

Input P1 Connector								Output					
19	20	24	27	28	52	59	68	J1-13	J1-17	J1-30	J1-37	J2-20	P1-12
H			L						L				
L			L						H				
L			H						H				
H			H						H				
H	L	H	L			L					L		
L	L	H	L			L					H		
H	L	H	L			H					H		
H	H	H	L			L					H		
H	L	L	L			L					H		
		L	L	L			L	L					
		H	L	L			L	H					
		L	H	L			L	H					
		L	L	H			H	H					
			L		L								L
			L		H								H
			H		L								H
			H		H								H
L		L	L	L			L			L			
H		L	L	L			L			H			
L		L	L	H			H			H			
H		L	L	L		L						L	
H		H	L	L		L						H	
H		L	L	H		L						H	
H		L	L	L		H						H	

Notes:
H = logic high
L = logic low

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3.5.6 Read/write decoders.

3.5.6.1 Read decoder operation. With power applied to the IOB per 3.5.2 and the inputs of Table III applied to the IOB, the outputs of READ decoder U55 shall be decoded low as shown in Table III. With I/O BOARD 1 SELECT, LDS and READ (P1-27, 28 & 24) logic low, data bus buffer U56 shall set to the "read" direction and allow data to flow from D7 through DO to B1/2 D7 through B1/2 DO. With I/O BOARD 1 SELECT, UDS and READ (P1-27, 68 & 24) logic low, data bus buffer U58 shall set to the "read" direction and allow data to flow from D15 through D8 to B1/2 D15 through B1/2 D8.

3.5.6.2 Write decoder operation. With power applied to the IOB per 3.5.2 and the inputs of Table IV applied to the IOB, the outputs of write decoder U52 shall be decoded low as shown in Table IV. With I/O BOARD 1 SELECT and LDS (P1-27 and 28) logic low and READ (P1-24) logic high, data bus buffer U56 shall set to the "write" direction and allow data to flow from B1/2 D7 through B1/2 D0 to D7 through D0. With I/O BOARD 1 SELECT and UDS (P1-27 and 68) logic low and READ (P1-24) logic high, data bus buffer U58 shall set to the "write" direction and allow data to flow from B1/2 D15 through B1/2 D8 to D15 through D8.

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TABLE III. Read decoder logic.

Input P1 Connector									Output Signal Low
27	24	64	28	19	59	20	60	21	
L	L	H	L	H	H	L	L	L	J2-1
L	L	H	L	H	H	L	L	H	J2-25
L	L	H	L	H	H	L	H	L	J1-29
L	L	H	L	H	H	L	H	H	J1-26
L	L	H	L	H	H	H	L	L	J1-3
L	L	H	L	H	H	H	L	H	J1-15
L	L	H	L	H	H	H	H	L	J1-1
L	L	H	L	H	H	H	H	H	J1-27

Notes:
H = logic high
L = logic low

TABLE IV. Write decoder logic.

Input P1 Connector									Output Signal Low
27	24	64	28	19	59	20	60	21	
L	H	L	L	H	H	L	L	H	J1-16
L	H	L	L	H	H	L	H	L	J1-4
L	H	L	L	H	H	H	L	H	J1-6
L	H	L	L	H	H	H	H	L	J1-2
L	H	L	L	H	H	H	H	H	J1-28

Notes:
H = logic high
L = logic low

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3.5.7 End-Around-Test (EAT) of bus 1/2 data and address lines.

3.5.7.1 EAT of bus 1/2 data lines. Writing a test word onto data bus lines B1/2 D0 through B1/2 D15 and decoding DATA LATCH WRITE (J1-4) low momentarily per 3.5.6.2 with UDS (P1-68) and LDS (P1-28) both low shall cause data bus latches U53 and U60 to latch the data bus data. The outputs of latches U53 and U60 shall be in a high-impedance state. After clearing the data lines, decoding DATA LATCH READ (J1-29) low per 3.5.6.1 shall force the outputs of the latches out of the high-impedance state and put the latched data onto the data bus. The data word read from the IOB at B1/2 D0 through B1/2 D15 shall agree with the original test word written to the IOB.

3.5.7.2 EAT of bus 1/2 address lines. With I/O BOARD 1 SELECT (P1-27), LDS (P1-28) and READ (P1-24) low, writing test bits to address lines B1/2 A1 through B1/2 A6 and decoding A TEST (J1-30) low per 3.5.5 shall cause the test bits to appear on data bus lines B1/2 D0 through B1/2 D7 as follows:

Bit A1 shall be on B1/2 D0, B1/2 D1
Bit A2 shall be on B1/2 D2
Bit A3 shall be on B1/2 D3
Bit A4 shall be on B1/2 D4
Bit A5 shall be on B1/2 D5
Bit A6 shall be on B1/2 D6, B1/2 D7

3.5.8 Analog to Digital (A/D) conversion initiation and UART reset operation. With power applied to the IOB per 3.5.2, TEST RESET (J1-22) high and STROBES (J1-37) decoded low per 3.5.5, decoding address lines B1/2 A1 and B1/2 A2 per table V shall cause the convert signal for the analog to digital converter U20 to go logic low or a reset signal for one of the three UARTs U42, U46 or U54 to go logic high per Table V. (NOTE: After resetting a UART, that UART should not be used for at least 40 microseconds or until its Transmitter Register Empty (TRE) line goes logic high.)

TABLE V. Convert and reset signals.

Address Lines		
P1-61	P1-21	OUTPUT SIGNAL
L	L	J2-2 LOW
L	H	J-11 HIGH
H	L	J1-24 HIGH
H	H	J1-23 HIGH

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3.5.9 TEST RESET operation. With power applied to the IOB per 3.5.2, pulsing TEST RESET (J1-22) logic low momentarily shall cause the logic levels of table VI for resetting all three on-board UARTs U42, U46 and U54 simultaneously, initializing the A/D converter U20, initializing the Unit 4 BIT circuit and clearing lamp driver latch U25.

TABLE VI. TEST RESET functions.

Test Point	Logic Level
J1-11	H (pulse)
J1-24	H (pulse)
J1-23	H (pulse)
J2-2	H (steady state)
J1-12	H (steady state)
J2-5	L (steady state)
J2-6	L (steady state)
J2-7	L (steady state)
J2-8	L (steady state)

3.5.10 BIT latch U35 and Transmit latch U59.

3.5.10.1 BIT latch U35 operation. With power applied to the IOB per 3.5.2 and with BIT LATCH WRITE decoded low momentarily (P1-21, 27, 28, 60 and 64 set logic low & P1-19, 20, 24 and 59 set logic high) data bits B1/2 D0 through B1/2 D7 shall latch onto the output control lines of latch U35 as follows:

Data Bus Input to IOB	Latched Onto Output Control Line	Test Point
B1/2 D7 (P1-5)	BIT MUX CONTROL C	J1-49
B1/2 D6 (P1-45)	BIT MUX CONTROL B	J1-48
B1/2 D5 (P1-6)	BIT MUX CONTROL A	J1-47
B1/2 D4 (P1-46)	PANEL EAT ENABLE	J1-19
B1/2 D3 (P1-7)	UNIT 4 EAT ENABLE	J1-18
B1/2 D2 (P1-47)	RANGE METER ENABLE	J1-50
B1/2 D1 (P1-8)	UNIT 4 MSB TEST ENABLE	J1-21
B1/2 D0 (P1-48)	NO CONNECTION	-

Latched BIT MUX CONTROL lines C, B and A shall set up multiplexer U29 and output decoder U28 to perform the functions shown in Table VII. The serial input to the BIT UART U46 shall have TTL logic levels at BIT RCV DATA (J1-31), except logic highs shall be pulled up by a resistor U67 to exceed the 3.0 V CMOS threshold of the UART.

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TABLE VII. BIT MUX control line decoding.

BIT MUX Control Line			Decoded Function
C (J1-49)	B (J1-48)	A (J1-47)	
L	L	L	BIT UART does not receive or transmit data anywhere
L	L	H	DIST EAT DATA
L	H	L	XMIT 5 EAT DATA
L	H	H	XMIT 2 EAT DATA
H	L	L	BIT EAT DATA
H	L	H	Transmit to CPU TO DISTRIBUTION BOX DATA; P2-8(HI), P2-43 (LO) Receive from DISTRIBUTION BOX TO CPU DATA; P2-3(HI), P2-38(LO)
H	H	L	Transmit to CPU TO UNIT 5 DATA; P2-9 (HI), P2-44 (LO) Receive from UNIT 5 TO CPU DATA; P2-4 (HI), P2-39 (LO)
H	H	H	Transmit to CPU TO UNIT 2 DATA; P2-10 (HI), P2-45 (LO) Receive from UNIT 2 TO CPU DATA; P2-5 (HI), P2-40 (LO)

NOTES:

H = logic high
L = logic low

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3.5.10.2 Transmit latch U59 operation. With power applied to the IOB per 3.5.2, data on upper data bus lines B1/2 D8 through B1/2 D15 shall latch onto data lines UD0 through UD7 at J1-35, 10, 34, 9, 33, 8, 32 and 7, respectively, when:

a. $\overline{\text{2 DELAY WRITE}}$ (P1-64) is logic low
b. $\overline{\text{UDS}}$ (P1-68) is logic low
c. $\overline{\text{I/O BOARD 1 SELECT}}$ (P1-27) is logic low
d. $\overline{\text{READ}}$ (P1-24) is logic high

3.5.11 BIT UART U46.

3.5.11.1 Normal BIT UART operation. BIT UART U46 shall transmit and receive data to and from the ports of Table VII with formats per Figures 2 and 3. Transmissions select the data channel of an analog voltage to be measured and start the analog to digital conversion at a remote BIT assembly. Received data is that of the converted data in binary offset digital format.

3.5.11.1.1 BIT UART transmitting requirements.

a. Power and clock shall be applied to the IOB per 3.5.2 and 3.5.4, respectively.

b. BIT UART U46 shall be reset per 3.5.8.

c. The Transmitter Buffer and Transmitter Registers of BIT UART U46 shall be empty to enable two bytes of data to be loaded for transmission; i.e., TBRE (B1/2 D9) and TRE (B1/2 D8) shall both be logic high after a reset as seen by decoding $\overline{\text{BIT UART STATUS READ}}$ (J1-3) low per 3.5.6.1 with $\overline{\text{UDS}}$ (P1-68) low.

d. The IOB output (and input) port selection per Table VII shall be input onto the lower byte of the data bus with bits B1/2 D4 through B1/2 D0 as "don't cares."

e. The port selection shall be latched by pulsing $\overline{\text{BIT LATCH WRITE}}$ low (P1-21, 27, 28, 60 and 64 set logic low and P1-19, 20, 24 and 59 set logic high).

f. The channel number for the remote BIT analog voltage to be measured shall be input onto the upper byte of the data bus with bit B1/2 D15 low.

g. The channel number shall be latched onto UD7 through UD0 per 3.5.10.2.

h. The channel number shall be loaded into UART U46 and transmitted by pulsing $\overline{\text{BIT UART LOAD}}$ (J1-28) low per 3.5.6.2.

i. The channel number shall be transmitted again, except with the B1/2 D15 bit high instead of low to start the A/D conversion on the remote BIT assembly.

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3.5.11.1.2 BIT UART receiving requirements.

a. Power and clock shall be applied to the IOB per 3.5.2 and 3.5.4, respectively.

b. RS-422 differential serial data shall be clocked into the IOB at the port selected in 3.5.11.1.1.d and 3.5.11.1.1.e and have TTL voltage levels at BIT RCV DATA (J1-31), except resistor U67 shall pull up logic highs to exceed the 3.0 volt CMOS threshold of the UART.

c. BIT UART status bits shall be read by decoding $\overline{\text{BIT UART STATUS READ}}$ (J1-3) low per 3.5.6.1 with $\overline{\text{UDS}}$ (P1-68) low to determine when a byte of data has been received ($\overline{\text{DR}}$ (J1-12), B1/2 D10 shall be high). The status bits shall also be read to ensure there are no parity, overrun or framing errors (B1/2 D11, B1/2 D12 and B1/2 D13 shall be low).

d. BIT UART Receiver Buffer Register lines shall exit the high-impedance state and load the received data byte onto the upper byte of the data bus when $\overline{\text{BIT UART READ}}$ (J1-27) is decoded low per 3.5.6.1 with $\overline{\text{UDS}}$ (P1-68) low.

3.5.11.2 BIT UART end-around-testing. BIT UART U46 shall perform end-around-testing in the same manner as in 3.5.11.1.1 and 3.5.11.1.2, except EAT loops shall be selected per Table VII in lieu of remote BIT assembly ports, and test bytes shall be transmitted in lieu of channel numbers. The received bytes shall agree with the test bytes transmitted.

3.5.12 Panel UART U54.

3.5.12.1 Normal Panel UART operation. Panel UART U54 shall transmit serial data to IOB output port CPU TO CONTROL PANEL DATA at P2-7(HI) and P2-42(LO) and receive serial data from IOB input port CONTROL PANEL TO CPU DATA at P2-2(HI) and P2-37(LO) with data format per Figure 3. Transmissions request weapon system Ballistic/BIT Panel switch status and set the value of the display digits on the same panel. Received data is the switch status information.

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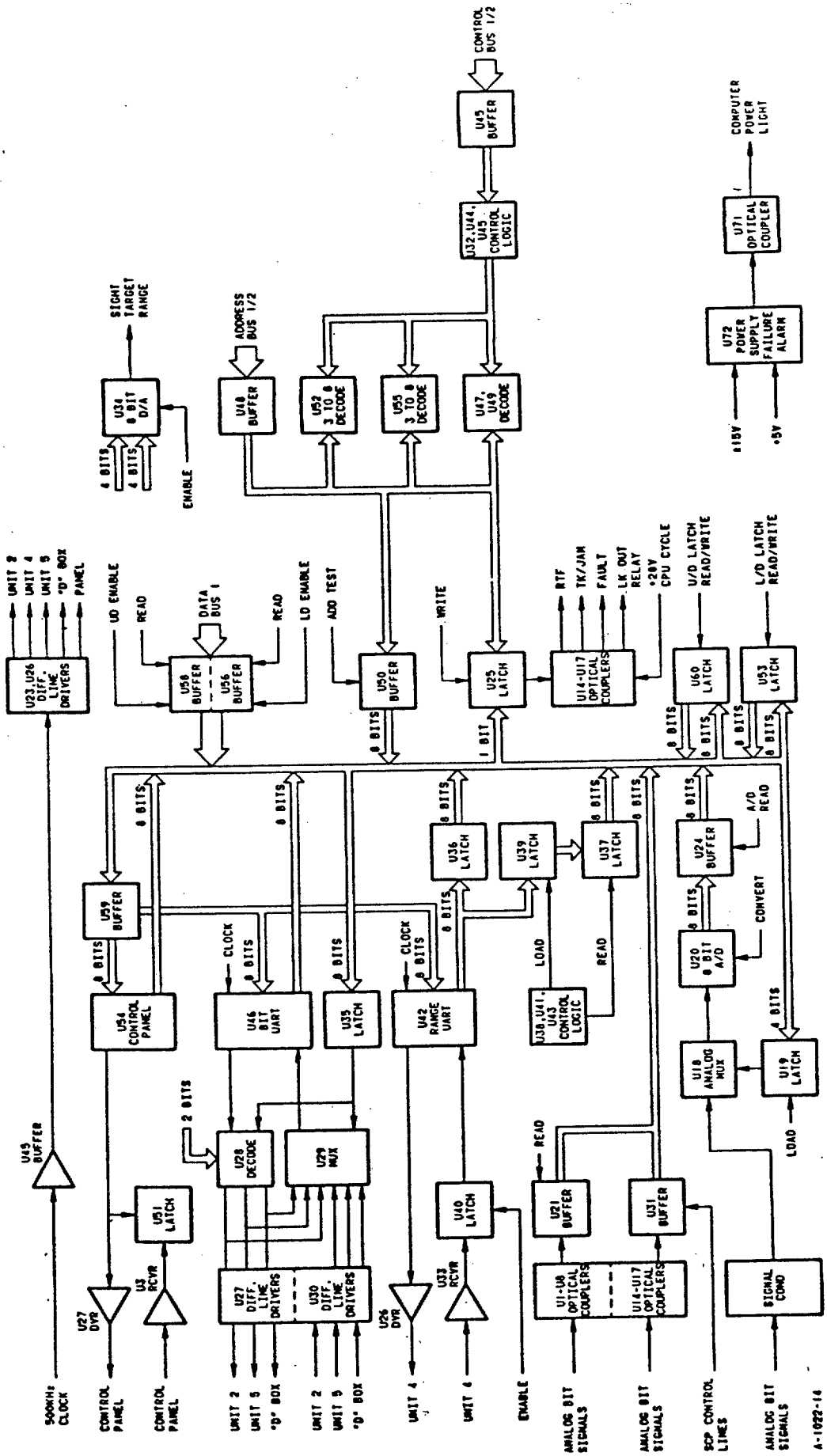
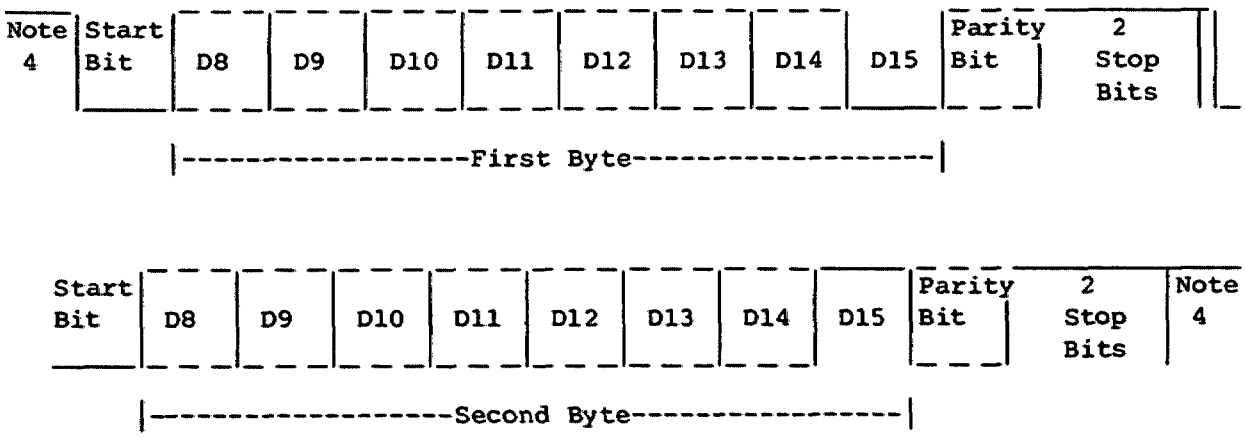


Figure 1. IOB Functional Block Diagram

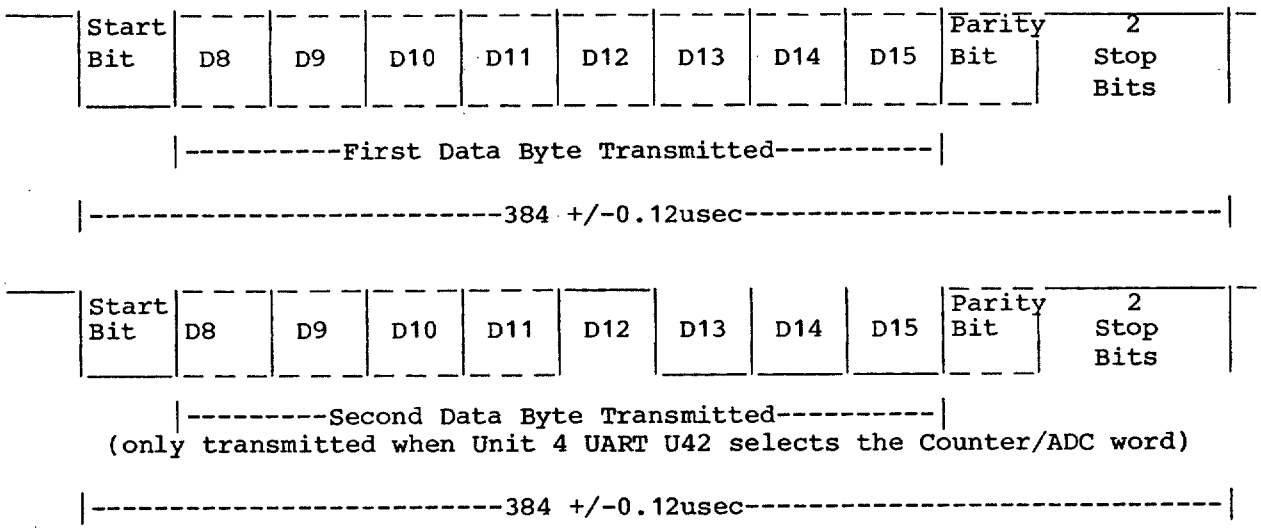
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NOTES:
1. Start bits are always logic low and stop bits are always logic high.
2. First byte transmitted has D15 logic low to call up analog channel and second byte transmitted has D15 logic high to start A/D conversion.
3. Parity bits for both bytes are odd.
4. The data line is logic high when no data is being transmitted.
5. The data frequency shall be 31.250kHz +/-5Hz.

FIGURE 2. BIT UART U46 transmit data format.

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NOTES:

1. The data line shall be logic high when no data is being transmitted or received.
2. The start bits are always logic low and the stop bits are always logic high.
3. For BIT UART U46 and Panel UART U54, the parity bit is odd. For Unit 4 UART U42 the parity bit(s) are "don't-care(s)", because the parity error flag output on the Unit 4 UART in the Radar Range Computer is not connected.
4. The second byte is only transmitted if the Counter/ADC word is selected.
5. The D12 bit is logic high in the second byte transmitted to reset the digital counter and analog to digital converter within Unit 4.
6. The data frequency shall be 31.250 kHz +/-5 Hz.

FIGURE 3. BIT UART U46 receive, Panel UART U54 transmit/receive and Unit 4 UART U42 transmit data formats

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3.5.12.1.1 Panel UART transmitting requirements.

a. Power and clock shall be applied to the IOB per 3.5.2 and 3.5.4, respectively.

b. Panel UART U54 shall be reset per 3.5.8.

c. The Transmitter Buffer and Transmitter Registers of Panel UART U54 shall be empty to enable a byte of data to be loaded for transmission; i.e. TBRE (B1/2 D8) and TRE (B1/2 D9) shall both be logic high after a reset as seen by decoding PANEL STATUS READ (J1-26) low per 3.5.6.1 with UDS (P1-68) low.

d. A data byte for transmission shall be input onto the upper byte of the data bus.

e. The data byte shall be latched onto UD7 through UDO per 3.5.10.2.

f. The data byte shall be loaded into Panel UART U54 and transmitted by pulsing PANEL UART LOAD (J1-2) low per 3.5.6.2.

3.5.12.1.2 Panel UART receiving requirements.

a. Power and clock shall be applied to the IOB per 3.5.2 and 3.5.4, respectively.

b. BIT latch U35 shall be set up to enable Panel UART U54 to receive data from IOB input port CONTROL PANEL TO CPU DATA (P2-2 and 37) by inputting and latching B1/2 D4 logic low (with all other low byte bits as "don't cares") per 3.5.10.1. PANEL EAT ENABLE (J1-19) shall be logic low.

c. Clocked-in RS-422 differential serial data shall be converted to TTL levels at PANEL RCV DATA (J1-5), except resistor U69 shall pull up logic highs to exceed the 3.0 volt CMOS threshold of the UART.

d. Panel UART status bits shall be read by decoding PANEL STATUS READ (J1-26) low per 3.5.6.1 with UDS (P1-68) low to determine when a byte of data has been received (DR (J1-12), B1/2 D10 shall be high). The status bits shall also be read to ensure there are no parity, overrun or framing errors (B1/2 D11, B1/2 D12 and B1/2 D13 shall be low).

e. Panel UART Receiver Buffer Register lines shall exit the high-impedance state and load the received data byte onto the upper byte of the data bus when PANEL UART READ (J1-1)is decoded low per 3.5.6.1 with UDS (P1-68) low.

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3.5.12.2 Panel UART end-around-test. Panel UART U54 shall perform end-around-testing in the same manner as in 3.5.12.1.1 and 3.5.12.1.2, except the data byte for transmission shall be a test byte and bit B1/2 D4 shall be logic high instead of low to configure the UART to receive its own transmitted data (PANEL EAT ENABLE (J1-19) logic high). Each received byte shall agree with each test byte transmitted.

3.5.13 Unit 4 UART U42.

3.5.13.1 Normal Unit 4 UART operation. Unit 4 UART U42 shall serially transmit RS-422 differential data to the weapon system radar range computer (Unit 4) through IOB output port CPU TO UNIT 4 DATA with P2-11(HI) and P2-46(LO). The transmitted data shall request continuous range information, samples of digital or analog test point voltages within Unit 4 or act as software test switches to override the hardware test switches within Unit 4. Unit 4 UART U42 shall serially receive RS-422 differential data through IOB input port UNIT 4 TO CPU DATA with P2-6(HI) and P2-41(LO). Transmit data format shall be per Figure 3 and receive data format per Figure 4. The lower data byte of each received word shall arrive first, followed by its related upper byte. The lower byte shall have an odd parity bit, which shall generate the low to high transition of BYTE LOAD (J1-20) and latch the lower byte while the upper byte is being received. When the upper byte with even parity is received, WORD LOAD (J1-25) shall transition from low to high and latch the lower and upper bytes onto data lines B1/2 D15 through B1/2 D0.

3.5.13.1.1 Unit 4 UART transmitting requirements.

a. Power and clock shall be applied to the IOB per 3.5.2 and 3.5.4, respectively.

b. Unit 4 UART U42 shall be reset per 3.5.8.

c. The Transmitter Buffer and Transmitter Registers of Unit 4 UART U42 shall be empty to enable a byte of data to be loaded for transmission; i.e. TBRE (B1/2 D15) and TRE (B1/2 D14) shall both be logic high after a reset as seen by decoding BIT UART STATUS READ (J1-3) low per 3.5.6.1 with $\overline{\text{UDS}}$ (P1-68) low. When the Counter/ADC word is selected, only the Transmitter Buffer Register must be empty before loading the second byte for transmission.

d. Each data byte for transmission shall be input onto the upper byte of the data bus.

e. Each data byte for transmission shall be latched onto UD7 through UDO per 3.5.10.2.

f. Each data byte for transmission shall be loaded into Unit 4 UART U42 and transmitted by pulsing $\overline{\text{UNIT 4 UART LOAD}}$ (J1-6) low per 3.5.6.2.

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3.5.13.1.2 Unit 4 UART receiving requirements.

a. Power and clock shall be applied to the IOB per 3.5.2 and 3.5.4, respectively.

b. BIT latch U35 shall be set up to enable Unit 4 UART U42 to receive data from IOB input port UNIT 4 TO CPU DATA (P2-6 and 41), by inputting and latching B1/2 D3 logic low (with B1/2 D1 low and all other low byte bits as "don't cares") per 3.5.10.1. UNIT 4 EAT ENABLE (J1-18) and UNIT 4 MSB TEST ENABLE (J1-21) shall both be logic low.

c. Clocked-in RS-422 differential serial data per Figure 4 shall be converted to TTL levels at UNIT 4 RCV DATA (J1-14), except resistor U66 shall pull up logic highs to exceed the 3.0 volt CMOS threshold of the UART.

d. The first byte of each word received by Unit 4 UART U42 shall generate BYTE LOAD (J1-20), which shall latch this first byte into latch U39. The second byte of each word received shall generate WORD LOAD (J1-25), which shall latch the second byte into latch U36 and the first byte into latch U37. Decoding $\overline{\text{UNIT 4 READ}}$ (J1-15) low per 3.5.6.1 with $\overline{\text{UDS}}$ (P1-68) logic low shall cause the output lines of latches U36 and U37 to exit the high-impedance state and load the received data word onto data bus lines B1/2 D15 through B1/2 D0.

3.5.13.2 Unit 4 UART end-around-test requirements.

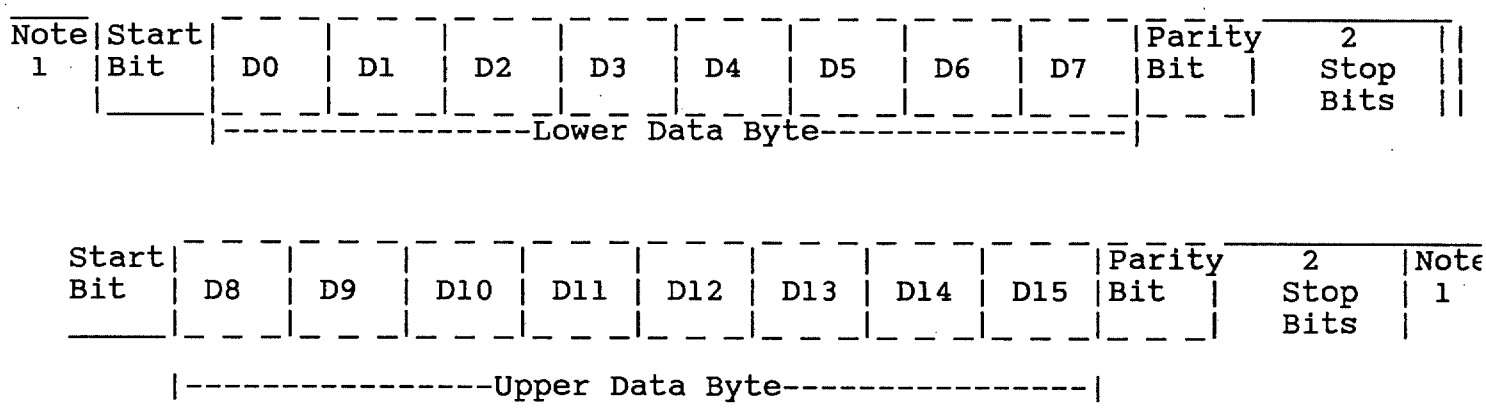
a. End-around-test shall be set up by inputting B1/2 D1 and B1/2 D3 logic high with all other low data byte bits as "don't cares," then pulsing $\overline{\text{BIT LATCH WRITE}}$ low (P1-21, 27, 28, 60 and 64 set logic low and P1-19, 20, 24 and 59 set logic high) to latch UNIT 4 EAT ENABLE (J1-18) and UNIT 4 MSB TEST ENABLE (J1-21) high.

b. Test bytes loaded onto upper data bus lines B1/2 D15 through B1/2 D8 and transmitted per 3.5.13.1.1 shall wrap around and be received by Unit 4 UART U42.

c. The logic high on UNIT 4 MSB TEST ENABLE shall hold BYTE LOAD (J1-20) high and cause WORD LOAD (J1-25) to pulse low for every byte received. WORD LOAD shall latch each received byte into the upper data byte latch U36. The high on BYTE LOAD shall prevent received bytes from latching into lower data byte latch U39.

d. Decoding $\overline{\text{UNIT 4 READ}}$ (J1-15) low per 3.5.6.1 with $\overline{\text{UDS}}$ (P1-68) low for each received byte shall dump the received byte onto upper data bus lines B1/2 D15 through B1/2 D8, where it shall compare exactly to the corresponding transmitted test byte.

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NOTES:
1. The data line shall be logic high when no data is being received.
2. Start bits are always logic low and stop bits are always logic high.
3. The data frequency shall be 31.250kHz +/-5Hz.
4. The first byte of each data word received shall be the lower byte and hav odd parity. The second byte of each data word received shall be the uppe byte and have even parity.

FIGURE 4. Unit 4 UART U42 receive data format.

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3.5.14 Range meter output. Unit 4 UART U42 shall convert received digital serial range words to parallel and digital to analog converter U34, when enabled, shall convert 8 bits of the parallel words to analog voltages for output from the IOB to drive the range meter in the weapon system sight.

3.5.14.1 Range meter output requirements.

a. Power and clock shall be applied to the IOB per 3.5.2 and 3.5.4, respectively.

b. Unit 4 UART U42 shall be reset per 3.5.8.

c. $\overline{\text{RANGE METER ENABLE}}$ (J1-50) shall be latched low by inputting a low on B1/2 D2 (with B1/2 D1 and B1/2 D3 low and all other low byte data bits "don't cares"), then pulsing $\overline{\text{BIT LATCH WRITE}}$ low (P1-21, 27, 28, 60 and 64 set logic low and P1-19, 20, 24 and 59 set logic high).

d. A serial word per Figure 4 shall be clocked into UNIT 4 TO CPU DATA (P2-6 and 41) with data bits D11 through D4 per Table VIII and all other data bits as "don't cares."

e. The analog range voltage (A2-1 (HI) and P2-36 (LO)) output shall be per Table VIII for each digital serial input code of Table VIII.

Table VIII. Range voltage D/A conversion.

Digital Serial Input D11 through D4	Range Voltage Output P2-1(HI) P2-36(LO)
Binary	Allowable tolerance: ± 0.0975 Vdc
0000 0000	0
0000 0001	0.039 Vdc
0000 0010	0.078 Vdc
0000 1111	0.586 Vdc
0001 0000	0.625 Vdc
0111 1111	4.961 Vdc
1000 0000	5.000 Vdc
1100 0000	7.500 Vdc
1111 1111	9.961 Vdc

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3.5.15 Analog multiplexer and analog to digital (A/D) converter operation. With the A/D converter initialized per 3.5.9 and the analog voltages (Vin) applied per Table IX, each analog input voltage shall be selected and converted to digital output data, when the following requirements are met.

a. An IOB channel address shall be loaded onto data bus lines B1/2 D3 through B1/2 D0.

b. Decoding $\overline{\text{ANALOG CHAN LOAD}}$ low (P1-20, 27, 28 and 64 set logic low and P1-19, 21, 24, 59 and 60 set logic high shall latch the data bus address onto multiplexer address lines AMUX 3 through AMUX 0 per Table IX.

c. The multiplexer U18 shall output the selected channel at A to D IN (J2-4) with the scaled voltage level per Table X.

d. $\overline{\text{CONVERT}}$ (J2-2) shall be decoded low per 3.5.8 to start the analog to digital conversion.

e. The $\overline{\text{A DATA READY}}$ signal shall go low when the A/D conversion is complete. Decoding $\overline{\text{SWITCH}}$ INPUT 2 READ (J2-20) low per 3.5.5 shall allow the $\overline{\text{A DATA READY}}$ signal to be available on B1/2 D7 as a high. The conversion shall require 40 microseconds maximum.

f. Decoding $\overline{\text{A TO D READ}}$ (J2-1) low per 3.5.6.1 shall put the offset binary output of the A/D converter onto data bus lines B1/2 D7 through B1/2 D0. The digital output shall be within the range specified in Table X.

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TABLE IX. Analog multiplexer inputs.

Mux Input	IOB Channel Address AMUX 3 (J2-24)	AMUX 2 (J2-23)	AMUX 1 (J2-22)	AMUX 0 (J2-21)	Analog Input/ Connector-Pin	Analog Input Vin (in Vdc unless otherwise stated)
S1	0	0	0	0	GUN COMMAND EL/P2-58(HI), P2-57(LO)	-0.50+0.15
S1	0	0	0	0	GUN COMMAND EL/P2-58(HI), P2-57(LO)	+0.50+0.15
S2	0	0	0	1	GUN COMMAND AZ/P2-23(HI), P2-22(LO)	-2.5+0.5
S2	0	0	0	1	GUN COMMAND AZ/P2-23(HI), P2-22(LO)	+2.5+0.5
S3	0	0	1	0	115VAC 400HZ REF/P2-70(R2),P2-35(R1) NOTE 1	115+6VACrms 400 HZ
S4	0	0	1	1	RANGE VOLTAGE/per 3.5.14	FF (hexadecimal)
S5	0	1	0	0	+24V/P2-21, P2-56 RTN NOTE 2	+24.0+1.2
S6	0	1	0	1	-24V/P2-55, P2-56 RTN NOTE 2	-24.0+1.2
S7	0	1	1	0	MANUAL SPEED/P2-62; P1-38, 78 RTN	+3.8+0.3
S8	0	1	1	1	-15V/INTERNAL	-15.00+0.75
S9	1	0	0	0	+15V/INTERNAL	+15.00+0.75
S10	1	0	0	1	AZ MOT 1 CW/P2-24, P1-38,78 RTN	+24.0+1.2
S11	1	0	1	0	AZ MOT 1 CCW/P2-59, P1-38, 78 RTN	+24.0+1.2
S12	1	0	1	1	AZ MOT 2 CW/P2-25, P1-38, 78 RTN	+24.0+1.2
S13	1	1	0	0	AZ MOT 2 CCW/P2-60, P1-38, 78 RTN	+24.0+1.2
S14	1	1	0	1	EL MOT UP/P2-26, P1-38,78 RTN	+24.0+1.2
S15	1	1	1	0	EL MOT DOWN/P2-61, P1-38, 78 RTN	+24.0+1.2
S16	1	1	1	1	+5V/INTERNAL	+5.00+0.25

NOTES:

1. P2-70 must be connected to P1-38, 78.

2. P2-56 must be connected to P1-38, 78.

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TABLE X. A/D inputs and outputs

MUX Input	IOB Channel Address AMUX 3 (J2-24) B1/2 D3	AMUX 2 (J2-23) B1/2 D2	AMUX 1 (J2-22) B1/2 D1	AMUX 0 (J2-21) B1/2 D0	A TO D IN (J2-4) (in Vdc unless Otherwise stated)	A/D Output Word From D7----DO(Dec) (Binary)	To D7----DO(Dec) (Binary)
S1	0	0	0	0	-0.682 to -1.334	01101111	01011101
S1	0	0	0	0	+0.682 to +1.334	10010001	10100011
S2	0	0	0	1	-1.955 to -3.065	01001111	00110001
S2	0	0	0	1	+1.955 to +3.065	10110001	11001111
S3	0	0	1	0	+3.853 to +4.278 Vpk	10111101	11001100
S3	0	0	1	0	-3.853 to -4.278 Vpk	00011110	00010010
S4	0	0	1	1	+4.280 to +4.477	11101100	11110100
S5	0	1	0	0	+3.175 to +3.643	11010000	11011110
S6	0	1	0	1	-3.175 to -3.643	00110000	00100010
S7	0	1	1	0	+1.516 to +1.827	10100110	10101111
S8	0	1	1	1	-3.952 to -4.506	00011100	00001011
S9	1	0	0	0	+3.952 to +4.506	11100100	11110101
S10	1	0	0	1	+3.175 to +3.643	11010000	11011110
S11	1	0	1	0	+3.175 to +3.643	11010000	11011110
S12	1	0	1	1	+3.175 to +3.643	11010000	11011110
S13	1	1	0	0	+3.175 to +3.643	11010000	11011110
S14	1	1	0	1	+3.175 to +3.643	11010000	11011110
S15	1	1	1	0	+3.175 to +3.643	11010000	11011110
S16	1	1	1	1	+3.940 to +4.394	11100100	11110010

NOTES:

1. P2-70 must be connected to P1-38, 78.

2. P2-56 must be connected to P1-38, 78.

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3.5.16 Lamp and action-switch-lockout-relay drivers.

3.5.16.1 Latch operation. With latch U25 cleared per 3.5.9, setting LAMP WRITE low (P1-27, 59 and 64 set logic low and P1-19 and 20 set logic high) will enable U25 and allow data written to B1/2 D0 to be latched onto the U25 latch output selected by address lines A3, A2 and A1 per Table XI. The latched output shall control the logic level of the corresponding driver output line shown in Table XI.

TABLE XI. Latch U25 output decoding

Address A3	A2	A1	Latch Output	Driver Output Line
0	0	1	LATCH 0 (J2-5)	ACTION SWITCH LOCKOUT (P2-17)
0	1	0	LATCH 1 (J2-6)	FAULT LAMP (P2-52)
0	1	1	LATCH 2 (J2-7)	TRACK/JAM LAMP (P2-18)
1	0	0	LATCH 3 (J2-8)	READY TO FIRE LAMP (P2-53)

3.5.16.2 Driver outputs. IOB connector-pins P2-17, 18, 52 and 53 shall be connected through external pull-up resistors to +24 +6-2 Vdc for the following output requirements. The READY TO FIRE LAMP, TRACK/JAM LAMP and FAULT LAMP outputs shall be capable of sinking 40 milliamperes minimum. The ACTION SWITCH LOCKOUT output shall be capable of sinking 30 milliamperes minimum.

a. A logic high on any U25 latch output shall cause the corresponding driver output line to go low (0 ± 2 Vdc) to 24V (BAT) RTN (P2-19). A logic low on any U25 latch output shall cause the corresponding driver output line to go high (+24 +6-2 Vdc).

b. With LATCH 1 logic low, FAULT LAMP (P2-52) shall go logic low (0 ± 2 Vdc) when +24 +6-2 Vdc is applied to COMPUTER CYCLE at P2-54.

c. With LATCH 0 logic high, data bit B1/2 D4 shall go logic high when SWITCH INPUT 2 READ (J2-20) is decoded low per 3.5.5.

3.5.17 Switch inputs.

3.5.17.1 Switch 1 read operation. With power per 3.5.2 and input voltages per Table XII applied to the IOB, decoding SWITCH INPUT 1 READ (J2-25) low per 3.5.6.1, shall cause B1/2 D7 through B1/2 D0 to have the logic levels shown in Table XII.

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TABLE XII. Switch 1 read.

Switch Inputs								Data Bus Status							
FAULT BYPASS P2-65	EXT. RANGE ENABLE P2-30	TEST MODE P2-66	GROUND MODE P2-31	RADAR MODE P2-67	MANUAL MODE P2-32	EXT. MODE P2-68	FAN SENSOR INPUT P2-33	B1/2 D7 P1-5	B1/2 D6 P1-45	B1/2 D5 P1-6	B1/2 D4 P1-46	B1/2 D3 P1-7	B1/2 D2 P1-47	B1/2 D1 P1-8	B1/2 D0 P1-48
+24 +6/-2 Vdc	+24 +6/-2 Vdc	+24 +6/-2 Vdc	+24 +6/-2 Vdc	+24 +6/-2 Vdc	+24 +6/-2 Vdc	+24 +6/-2 Vdc	0 to +12 +/-4 V pk., 215 +/-100 Hz sq. wave, 50% duty cycle	H	H	H	H	H	H	H	NOTE 3
0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	L	L	L	L	L	L	L	L

NOTES:
1. H = TTL logic high.
   L = TTL logic low.
2. The return for all input voltages is P2-27.
3. Logic level follows input square wave, i.e. peak = H, valley = L.

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3.5.17.2 Switch 2 read operation. With power per 3.5.2 and input voltages per Table XIII applied to the IOB, decoding SWITCH INPUT 2 READ (J2-20) low per 3.5.5 shall cause B1/2 D7 through B1/2 D0 to have the logic levels shown in Table XIII.

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TABLE XIII. Switch 2 read.

Switch Inputs								Data Bus Status							
A DATA READY	SCB ENABLE	I/O BOARD 2 IN	ACTION SWITCH LOCK-OUT	ACQ. TRACK SWITCH	FIRE TRIG-GER SWITCH	RAD. COM'D	ACTION SWITCH	B1/2 D7	B1/2 D6	B1/2 D5	B1/2 D4	B1/2 D3	B1/2 D2	B1/2 D1	B1/2 D0
	P1-22	P1-30	P2-17 NOTE 6	P2-63	P2-28	P2-64	P2-29	P1-5	P1-45	P1-6	P1-46	P1-7	P1-47	P1-8	P1-48
L (per 3.5. 15.5)	L	L	NOTE 4	+24 +6/-2 Vdc	+24 +6/-2 Vdc	+24 +6/-2 Vdc	+24 +6/-2 Vdc	H	NOTE 2	H	H	H	H	H	H
H	H	H	NOTE 5	0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	0 +/-1 Vdc	L	NOTE 2	L	L	L	L	L	L

NOTES:
1. H = TTL logic high.
   L = TTL logic low.
2. Application of a logic low to P1-22 with $\overline{\text{SCP TEST CLEAR}}$ (J1-16) high shall produce a logic high on B1/2 D6. B1/2 D6 shall remain high when P1-22 is returned to logic high. Application of a logic low to $\overline{\text{SCP TEST CLEAR}}$ per 3.5.6.2 with P1-22 at logic high shall produce a logic low on B1/2 D6. B1/2 D6 shall remain low when $\overline{\text{SCP TEST CLEAR}}$ is returned to logic high.
3. The return for the +24 Vdc inputs is P2-27.
4. When LATCH 0 (J2-5) is logic high per 3.5.16.2.3.
5. When LATCH 0 (J2-5) is logic low per 3.5.16.2.
6. P2-17 shall be connected through a 560 ohm resistor to +24 +6/-2 Vdc.

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3.5.18 Input/output requirements. Input/output requirements shall be as specified in Tables XIV and XV. Transistor-Transistor Logic (TTL) signals are denoted as such under descriptive data in Tables XIV and XV. The minimum and maximum input and output TTL voltage levels shall be as follows:

	MIN	TYP	MAX	
VOL	-	0.2	0.4	Voltage Output Low
VOH	2.4	3.4	-	Voltage Output High
VIL	-	-	0.7	Voltage Input Low
VIH	2.0	-	-	Voltage Input High

TABLE XIV. Inputs

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-1	B1/2 D15	TTL
P1-2	B1/2 D13	TTL
P1-3	B1/2 D11	TTL
P1-4	B1/2 D9	TTL
P1-5	B1/2 D7	TTL
P1-6	B1/2 D5	TTL
P1-7	B1/2 D3	TTL
P1-8	B1/2 D1	TTL
P1-9	+5V RET	EQUIPMENT GROUND
P1-18	+5V RET	EQUIPMENT GROUND
P1-19	B1/2 A6	TTL
P1-20	B1/2 A4	TTL
P1-21	B1/2 A2	TTL
P1-22	$\overline{\text{SCB ENABLE}}$	TTL
P1-24	$\overline{\text{READ}}$	TTL
P1-27	$\overline{\text{I/O BD 1 SELECT}}$	TTL

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TABLE XIV. Inputs. (continued)

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-28	$\overline{\text{LDS}}$	TTL
P1-30	$\overline{\text{I/O BD 2 IN}}$	TTL
P1-32	+5 V	+5 Vdc ± 0.25 Vdc
P1-33	+5 V	+5 Vdc ± 0.25 Vdc
P1-34	+5 V	+5 Vdc ± 0.25 Vdc
P1-35	+5 V RTN	EQUIPMENT GROUND
P1-36	+5 V RTN	EQUIPMENT GROUND
P1-37	+5 V RTN	EQUIPMENT GROUND
P1-38	±15 V RET	EQUIPMENT GROUND
P1-39	-15 V	-15 Vdc ± 0.75 Vdc
P1-40	+15 V	+15 Vdc ± 0.75 Vdc
P1-41	B1/2 D14	TTL
P1-42	B1/2 D12	TTL
P1-43	B1/2 D10	TTL
P1-44	B1/2 D8	TTL
P1-45	B1/2 D6	TTL
P1-46	B1/2 D4	TTL
P1-47	B1/2 D2	TTL
P1-48	B1/2 D0	TTL
P1-49	+5 V RET	EQUIPMENT GROUND
P1-50	CLOCK (500KHz)	TTL
P1-52	$\overline{\text{DELAY 2}}$	TTL
P1-58	+5 V RET	EQUIPMENT GROUND
P1-59	B1/2 A5	TTL
P1-60	B1/2 A3	TTL

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TABLE XIV. Inputs. (continued)

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-61	B1/2 A1	TTL
P1-64	$\overline{\text{2 DELAY WRITE}}$	TTL
P1-68	$\overline{\text{UDS}}$	TTL
P1-72	+5 V	+5 Vdc ± 0.25 Vdc
P1-73	+5 V	+5 Vdc ± 0.25 Vdc
P1-74	+5 V	+5 Vdc ± 0.25 Vdc
P1-75	+5 V RTN	EQUIPMENT GROUND
P1-76	+5 V RTN	EQUIPMENT GROUND
P1-77	+5 V RTN	EQUIPMENT GROUND
P1-78	± 15 V RET	EQUIPMENT GROUND
P1-79	-15 V	-15 Vdc ± 0.75 Vdc
P1-80	+15 V	+15 Vdc ± 0.75 Vdc
P2-2	CONTROL PANEL TO CPU DATA (HI)	DIFFERENTIAL INPUT
P2-3	DIST. BOX TO CPU DATA (HI)	DIFFERENTIAL INPUT
P2-4	UNIT 5 TO CPU DATA (HI)	DIFFERENTIAL INPUT
P2-5	UNIT 2 TO CPU DATA (HI)	DIFFERENTIAL INPUT
P2-6	UNIT 4 TO CPU DATA (HI)	DIFFERENTIAL INPUT
P2-17	$\overline{\text{ACTION SWITCH LOCKOUT}}$	SINK 30 mA @ 24 Vdc
P2-19	LAMP RTN	24 Vdc RTN
P2-21	+24 VOLTS	24 Vdc ± 2 Vdc

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TABLE XIV. Inputs. (continued)

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
P2-22	GUN COMMAND AZ (LO)	-10 To +10 Vdc
P2-23	GUN COMMAND AZ (HI)	-10 To +10 Vdc
P2-24	AZIMUTH MOTOR 1 CW	0-24 Vdc
P2-25	AZIMUTH MOTOR 2 CW	0-24 Vdc
P2-26	ELEVATION MOTOR UP	0-24 Vdc
P2-27	24 V (BATTERY) RTN	24 Vdc RTN
P2-28	FIRE TRIGGER SWITCH	0 OR 24 Vdc
P2-29	ACTION SWITCH	0 OR 24 Vdc
P2-30	EXTERNAL RANGE ENABLE	0 OR 24 Vdc
P2-31	GROUND MODE	0 OR 24 Vdc
P2-32	MANUAL MODE	0 OR 24 Vdc
P2-33	FAN SENSOR INPUT	12 V PEAK SQ. WAVE 217 Hz
P2-35	115 VAC R1 REF	115 $\pm$ 6 Vac 400 $\pm$ 0.4 Hz
P2-37	CONTROL PANEL TO CPU DATA (LO)	DIFFERENTIAL INPUT
P2-38	DIST. BOX TO CPU DATA (LO)	DIFFERENTIAL INPUT
P2-39	UNIT 5 TO CPU DATA (LO)	DIFFERENTIAL INPUT

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TABLE XIV. Inputs. (continued)

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
P2-40	UNIT 2 TO CPU DATA (LO)	DIFFERENTIAL INPUT
P2-41	UNIT 4 TO CPU DATA (LO)	DIFFERENTIAL INPUT
P2-55	-24 VOLTS	-24 Vdc ± 2 Vdc
P2-56	± 24 V RTN	24 Vdc RTN
P2-57	GUN COMMAND EL (LO)	-10 TO + 10 Vdc
P2-58	GUN COMMAND EL (HI)	-10 TO + 10 Vdc
P2-59	AZIMUTH MOTOR 1 CCW	0 - 24 Vdc
P2-60	AZIMUTH MOTOR 2 CCW	0 - 24 Vdc
P2-61	ELEVATION MOTOR DOWN	0 - 24 Vdc
P2-62	MANUAL SPEED	0 - 15 Vdc
P2-63	ACQUISITION/ TRACK SWITCH	0 OR 24 Vdc
P2-64	RADIATE COMMAND	0 OR 24 Vdc
P2-65	FAULT BYPASS	0 OR 24 Vdc
P2-66	TEST MODE	0 OR 24 Vdc
P2-67	RADAR MODE	0 OR 24 Vdc
P2-68	EXTERNAL MODE	0 OR 24 Vdc
P2-70	115 VAC NEUTRAL	NOT USED ON IOB

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TABLE XV. Outputs.

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
P1-11	I/O BOARD 1 IN	Signal Ground
P1-12	DTACK	TTL
P2-1	RANGE VOLTAGE (HI)	0 - 10 Vdc OUTPUT
P2-7	CPU TO CONTROL PANEL DATA (HI)	DIFFERENTIAL OUTPUT
P2-8	CPU TO DIST. BOX DATA (HI)	DIFFERENTIAL OUTPUT
P2-9	CPU TO UNIT 5 DATA (HI)	DIFFERENTIAL OUTPUT
P2-10	CPU TO UNIT 2 DATA (HI)	DIFFERENTIAL OUTPUT
P2-11	CPU TO UNIT 4 DATA (HI)	DIFFERENTIAL OUTPUT
P2-12	UNIT 4 CLOCK (HI)	DIFFERENTIAL OUTPUT
P2-13	CONTROL PANEL CLOCK (HI)	DIFFERENTIAL OUTPUT
P2-14	DIST. BOX CLOCK (HI)	DIFFERENTIAL OUTPUT
P2-15	UNIT 5 CLOCK (HI)	DIFFERENTIAL OUTPUT
P2-16	UNIT 2 CLOCK (HI)	DIFFERENTIAL OUTPUT
P2-18	TRACK/JAM LAMP	SINK 40 mA @ 24 Vdc
P2-20	COMPUTER POWER LAMP	SINK 40 mA @ 24 Vdc

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TABLE XV. Outputs. (continued)

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
P2-36	RANGE VOLTAGE (LO)	RANGE VOLTAGE RTN
P2-42	CPU TO CONTROL PANEL DATA (LO)	DIFFERENTIAL OUTPUT
P2-43	CPU TO DIST. BOX DATA (LO)	DIFFERENTIAL OUTPUT
P2-44	CPU TO UNIT 5 DATA (LO)	DIFFERENTIAL OUTPUT
P2-45	CPU TO UNIT 2 DATA (LO)	DIFFERENTIAL OUTPUT
P2-46	CPU TO UNIT 4 DATA (LO)	DIFFERENTIAL OUTPUT
P2-47	UNIT 4 CLOCK (LO)	DIFFERENTIAL OUTPUT
P2-48	CONTROL PANEL CLOCK (LO)	DIFFERENTIAL OUTPUT
P2-49	DIST. BOX CLOCK (LO)	DIFFERENTIAL OUTPUT
P2-50	UNIT 5 CLOCK (LO)	DIFFERENTIAL OUTPUT
P2-51	UNIT 2 CLOCK (LO)	DIFFERENTIAL OUTPUT
P2-52	$\overline{\text{FAULT LAMP}}$	SINK 40 mA @ 24 Vdc
P2-53	$\overline{\text{READY TO FIRE LAMP}}$	SINK 40 mA @ 24 Vdc
P2-54	$\overline{\text{COMPUTER CYCLE}}$	SINK 40 mA @ 24 Vdc

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3.5.19 Test points. The IOB shall have test points as specified in Table XVI below.

TABLE XVI. Test points.

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
J1-1	$\overline{\text{PANEL UART READ}}$	TTL
J1-2	$\overline{\text{PANEL UART LOAD}}$	TTL
J1-3	$\overline{\text{BIT UART STATUS READ}}$	TTL
J1-4	$\overline{\text{DATA LATCH WRITE}}$	TTL
J1-5	PANEL RCV DATA	TTL
J1-6	$\overline{\text{UNIT 4 UART LOAD}}$	TTL
J1-7	UD7	TTL
J1-8	UD5	TTL
J1-9	UD3	TTL
J1-10	UD1	TTL
J1-11	UNIT 4 UART RESET	TTL
J1-12	$\overline{\text{DR}}$	TTL
J1-13	$\overline{\text{PREAD}}$	TTL
J1-14	UNIT 4 RCV DATA	TTL
J1-15	$\overline{\text{UNIT 4 READ}}$	TTL
J1-16	$\overline{\text{SCP TEST CLEAR}}$	TTL
J1-17	$\overline{\text{DECODE ENABLE}}$	TTL

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TABLE XVI. Test points. (continued)

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
J1-18	UNIT 4 EAT ENABLE	TTL
J1-19	PANEL EAT ENABLE	TTL
J1-20	BYTE LOAD	TTL
J1-21	UNIT 4 MSB TEST ENABLE	TTL
J1-22	$\overline{\text{TEST RESET}}$	TTL
J1-23	BIT UART RESET	TTL
J1-24	PANEL UART RESET	TTL
J1-25	WORD LOAD	TTL
J1-26	$\overline{\text{PANEL STATUS READ}}$	TTL
J1-27	$\overline{\text{BIT UART READ}}$	TTL
J1-28	$\overline{\text{BIT UART LOAD}}$	TTL
J1-29	$\overline{\text{DATA LATCH READ}}$	TTL
J1-30	$\overline{\text{A TEST}}$	TTL
J1-31	BIT RCV DATA	TTL
J1-32	UD6	TTL
J1-33	UD4	TTL
J1-34	UD2	TTL
J1-35	UD0	TTL
J1-37	$\overline{\text{STROBES}}$	TTL
J1-47	BIT MUX CONTROL A	TTL

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TABLE XVI. Test points. (continued)

CONNECTOR-PIN	SIGNAL NAME	DESCRIPTIVE DATA
J1-48	BIT MUX CONTROL B	TTL
J1-49	BIT MUX CONTROL C	TTL
J1-50	$\overline{\text{RANGE METER ENABLE}}$	TTL
J2-1	$\overline{\text{A TO D READ}}$	TTL
J2-2	$\overline{\text{CONVERT}}$	TTL
J2-4	A TO D IN	-5 V TO +5 V
J2-5	LATCH 0	TTL
J2-6	LATCH 1	TTL
J2-7	LATCH 2	TTL
J2-8	LATCH 3	TTL
J2-20	$\overline{\text{SWITCH INPUT 2 READ}}$	TTL
J2-21	AMUX 0	TTL
J2-22	AMUX 1	TTL
J2-23	AMUX 2	TTL
J2-24	AMUX 3	TTL
J2-25	$\overline{\text{SWITCH INPUT 1 READ}}$	TTL

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3.5.20 Spares. The IOB shall have spares as specified in Table XVII.

TABLE XVII. Spares.

CONNECTOR-PIN
J1-36
J1-38 thru J1-46
J2-3
J2-9 thru J2-19
J2-26 thru J2-50
P1-10
P1-13 thru 17
P1-23, 25, 26
P1-29, 31, 51
P1-53 thru 57
P1-62, 63, 65
P1-66, 67, 69
P1-70, 71
P2-69

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90°F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Circuit Board, Input/Output	9360881	1

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product

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selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 and 4.5 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table XVIII shall be performed on a 100% basis.

TABLE XVIII. Functional tests.
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One IOB out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table XVIII and shall then meet the tests in Table XIX. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XIX. Special sampling.

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as

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to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2 the test equipment listed in Table XX (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7. These items shall be assembled at the AN/USM-465A test station prior to test execution. Proper operation of interface devices and test equipment shall be verified before using same to run diagnostic tests.

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TABLE XX. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Interface Device TPS Adapter Board A	9361380
2.	Cable Assembly W1 (Qty 2)	9397497
3.	Test Plug P3	9397498
4.	Test Plug P5	9397499
5.	Test Plug P4	9397500
6.	Digital Card Tester	AN/USM-465A
7.	ComCode Program Tape	Version 2.0
8.	Test Program Set Manual	9361335
9.	Test Program Set Tape	TPT9361335
10.	Oscilloscope	Input impedance of 1 Mohm or greater, single sweep capability, with probe. 0 to ± 100 Vdc and 20 MHz capability. Need large bandwidth to look at ripple spikes (ref. para. 3.5.2).

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The IOB shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the IOB shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

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a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the IOB shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating. This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The performance of the input/output board shall be tested to meet the requirements of 3.5 using the test equipment listed in Table XX. The performance testing shall be in accordance with the IOB Test Program Set Manual (TPS) drawing 9361335.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

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5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360881. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The input/output assembly is used in the fire control processor in the PIVADS self-propelled and towed vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Applicable stock number.

d. First article sample requirements.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certification of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control Processor (FCP)
Test Program Set (TPS)
Vulcan Air Defense System (VADS)

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6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is the initial public release of unlimited distribution.

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Preparing activity:
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(Project 1285-A778)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70568B (AR)	6 FEBRUARY 1992

3. DOCUMENT TITLE: CIRCUIT BOARD, INPUT/OUTPUT: 9360881

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

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